

Data Collection and Preprocessing Phase

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Project Title: Harnessing EC2, VPC & MobaXterm for Cutting-Edge Entity Hosting
Maximum Marks: 2 Marks

Data Quality Report Template

Data Source	Data Quality Issue	Severity	Resolution Plan
EC2 Instance Logs (AWS Console)	Missing log entries during instance reboot or shutdown	Moderate	Enable AWS CloudWatch Logs for continuous monitoring and automatic log collection
VPC Network Configuration	Misconfigured routing tables or security groups blocking traffic	High	Reconfigure VPC route tables, open required ports (HTTP/HTTPS/SSH) and apply least privilege rules
MobaXterm SSH Sessions	Session disconnection due to unstable SSH keys or timeout	Moderate	Use persistent SSH key authentication and adjust keep-alive settings in MobaXterm
Web Server Files (Apache/Nginx on EC2)	Missing or incorrectly deployed index.html or landing page files	High	Validate deployment path /var/www/html/ and ensure correct file permissions using chmod
Static Website Hosting Data	Image or asset loading issues due to incorrect paths	Low	Use absolute paths or S3-hosted assets for reliable loading across environments
Security Group Rules (AWS)	Over-permissive inbound/outbound rules causing security risks	High	Restrict access to specific IP ranges and required ports only (SSH:22, HTTP:80, HTTPS:443)
EC2 Resource Utilization	High CPU usage during peak traffic simulation	Moderate	Optimize instance type or enable auto-scaling for better performance